3.4 Network hardware

Candidates should be able to:	Notes and guidance
1 Understand that a computer needs a network interface card (NIC) to access a network	
2 Understand what is meant by, and the purpose of, a media access control (MAC) address, including its structure	• A network interface card is given a MAC address at the point of manufacture • MAC addresses are usually written as hexadecimal • MAC addresses are created using the manufacturer code and the serial code
3 (a) Understand what is meant by, and the purpose of, an internet protocol (IP) address (b) Understand that there are different types of IP address	• An IP address is allocated by the network and it can be static or dynamic • Including the characteristics of, and differences between, IPv4 and IPv6
4 Describe the role of a router in a network	• A router sends data to a specific destination on a network • A router can assign IP addresses • A router can connect a local network to the internet

4 Software

4.1 Types of software and interrupts

Candidates should be able to:	Notes and guidance
1 Describe the difference between system software and application software and provide examples of each	• System software provides the services that the computer requires, including operating system and utility software • Application software provides the services that the user requires
2 Describe the role and basic functions of an operating system	• Including: – managing files – handling interrupts – providing an interface – managing peripherals and drivers – managing memory – managing multitasking – providing a platform for running applications – providing system security – managing user accounts

4.1 Types of software and interrupts continued

Candidates should be able to:	Notes and guidance
3 Understand how hardware, firmware and an operating system are required to run applications software	• Applications are run on the operating system • The operating system is run on the firmware • The bootloader (firmware) is run on the hardware
4 Describe the role and operation of interrupts	• Including: – how an interrupt is generated – how it is handled using an interrupt service routine – what happens as a result of the interrupts • Software interrupts include division by zero and two processes trying to access the same memory location • Hardware interrupts include pressing a key on the keyboard and moving the mouse

4.2 Types of programming language, translators and integrated development environments (IDEs)

Candidates should be able to:	Notes and guidance
1 Explain what is meant by a high-level language and a low-level language, including the advantages and disadvantages of each	• Advantages and disadvantages include: – ease of reading and writing code, e.g. low-level is hard to read – ease of debugging code – machine independence – direct manipulation of hardware
2 Understand that assembly language is a form of low-level language that uses mnemonics, and that an assembler is needed to translate an assembly language program into machine code	
3 Describe the operation of a compiler and an interpreter, including how high-level language is translated by each and how errors are reported	• A compiler translates the whole code at once before executing it, producing an executable file • An interpreter translates and executes the code line-by-line • A compiler provides an error report for the whole code if errors are detected • An interpreter stops execution when an error is found

4.2 Types of programming language, translators and integrated development environments (IDEs) continued

Candidates should be able to:	Notes and guidance
4 Explain the advantages and disadvantages of a compiler and an interpreter	• Including an understanding that an interpreter is mostly used when developing a program and a compiler is used to translate the final program
5 Explain the role of an IDE in writing program code and the common functions IDEs provide	• Including: – code editors – run-time environment – translators – error diagnostics – auto-completion – auto-correction – prettyprint

5 The internet and its uses

5.1 The internet and the world wide web

Candidates should be able to:	Notes and guidance
1 Understand the difference between the internet and the world wide web	• The internet is the infrastructure • The world wide web is the collection of websites and web pages accessed using the internet
2 Understand what is meant by a uniform resource locator (URL)	• A URL is a text-based address for a web page; it can contain the protocol, the domain name and the web page/file name
3 Describe the purpose and operation of hypertext transfer protocol (HTTP) and hypertext transfer protocol secure (HTTPS)	
4 Explain the purpose and functions of a web browser	• The main purpose of a web browser is to render hypertext markup language (HTML) and display web pages • Functions include: – storing bookmarks and favourites – recording user history – allowing use of multiple tabs – storing cookies – providing navigation tools – providing an address bar